

Operator station - WinCC HMI

Plant overview

controller + trend

Sequence

Alarms

↓ commands, setpoints

HMI connection

↑ values, states, alarms (cyclic)

Shared data - global data blocks, structured with PLC data types

Measurements
values · signal status · limit bits

Alarm words
packed fault bits

Device data
commands · setpoints · states

↑ write values

↓ read values

↑ write alarms

↑ auto commands

↓ read commands

↑ write states

PLC program

① Input processing

[cyclic interrupt] [FC per Signal]

Read raw input values

Detect sensor / signal fault

Scale to engineering units

Check alarm limits

Startup OB: initialise measuring ranges/limits

② Control logic

[main cycle + cyclic interrupt]

Interlocks - safety conditions

Sequence - step chain

Flow controller - PI loop

Averaging - ring buffer

Alarm packing - fault bits → words

Error OBs: catch faults, bus failure → alarm

③ Device control

[main cycle] [FB + instance DB per Device]

On/off valve block

Control valve block

Pump drive block

Auto: commands from sequence

Manual: commands from HMI

Release bit: blocks run only after commissioning

↑ measured values

↑ commands setpoints

↑ inputs

↓ outputs

Process / field level

Sensors

Flow; level; pressure; temperature

Process

tank → pump → tank

Actuators

on/off; control valves; pumps